GOVTECH SINGAPORE

10 Years Tech for Public Good

Responsible AI

Agentic Evaluations

Workstream Overview

UNRESTRICTED

INTERNAL

1. KM principles
    a. Follow KM and Documentation Guidelines generally
    b. Keep maintaining this deck of slides
    c. Present this deck of slides during weekly sync ups / sprint reviews
    d. If you need to give a special presentation, then you should:
        i. Create a new deck of slides and extract content from this deck
        ii. Rename the new deck of slides as ***{YYYY-MM-DD} - {Meeting Title}*** (e.g. ***2026-01-19 - Fortnightly Sync***)



Executive Summary

As GovTech shifts from static LLM applications to autonomous, multi-step agentic systems, ***evaluation*** becomes the critical path for safe deployment.

This workstream proposes ***METIS, a unified, extensible agentic evaluation suite*** designed for Whole-of-Government adoption—enabling agency teams to build fast while ensuring AI safety, alignment, and operational reliability by default.

Motivation	1. **Shift to Agentic AI**: Autonomous multi-step systems introduce non-deterministic execution, state drift, and compounding errors. 2. **Eval Bottlenecks**: Traditional static LLM tests fail to evaluate complex agent trajectories, slowing safe deployment.
Vision	1. **Unified Product Suite**: A single, consistent front-facing product (METIS) delivering automated AI safety, guardrails, and compliance by default across Whole-of-Government.
Objectives	1. **Speed to Production:** Graduate agency AI pilots from sandbox to production in weeks instead of quarters. 2. **Automated Governance:** Provide trace scoring, step-error analytics, and audit-ready risk metrics.
Target Audience	1. **Application Developers (e.g., Desk)**: Building end-user agentic applications who need rapid CI/CD feedback loops and application-level success metrics. 2. **Orchestration Platforms (e.g., ORCA, ATLAS)**: Middleware platforms that can embed native evaluation hooks directly into their execution runtimes. 3. **Agent Infrastructure Teams (e.g., Harness)**: Core infrastructure providers who can offer standardized logging, tracing, and eval data pipelines at the foundational layer.

Motivation	1. **Accelerating Velocity Without Compounding Risk**: High-speed experimentation risks real-world operational harm without dedicated trajectory and task-completion evaluation. 2. **Standardizing Evaluation Across WOG**: Avoid building in silos - a unified methodology creates a shared language for agent performance, reliability, and safety across all public sector applications. 3. **Operationalizing Responsible AI at Scale**: Eliminate redundant engineering across agencies by centralizing core evaluation standards, governance checks, and safety baselines into a single, shared asset.
Vision	1. **Whole-of-Government Capability Enablement**: Enable safe, rapid adoption of autonomous agentic software across all agencies by making rigorous evaluation a seamless part of the software delivery lifecycle. 2. **WOG Center of Excellence & SME**: Position the Responsible AI team as the trusted, authoritative Subject Matter Expert for agentic evaluation, trajectory analysis, and safety benchmarks across government. 3. **SOTA Alignment & Open-Source Thought Leadership**: Maintain alignment with state-of-the-art (SOTA) academic research and actively drive thought leadership in the open-source community to shape international standards for trustworthy AI.
Objectives	**1. Build a Modular, Engine-Agnostic Evaluation Toolkit** a. Interoperable: API- and SDK-first architecture that seamlessly hooks into any agent framework or runtime environment. b. Customizable: Allows teams to write custom evaluators for domain-specific business logic, accuracy, tool invocation correctness, and trajectory efficiency. c. Extensible: Supports a plugin model for custom reporting outputs, telemetry destinations, and automated CI/CD pipeline triggers. 2. **Provide "Gov-Ready" Defaults**: Pre-configure default evaluator suites aligned with GovTech Responsible AI principles, security guardrails, policy compliance, and hallucination metrics out of the box. 3. **Drive Frictionless Ecosystem Adoption**: Accelerate onboarding through developer-centric CLI tools, interactive dashboards, reference templates, and hands-on technical enablement across agency engineering teams.
Target Audience	1. **Application Developers (e.g., Desk)**: Building end-user agentic applications who need rapid CI/CD feedback loops and application-level success metrics. 2. **Orchestration Platforms (e.g., ORCA, ATLAS)**: Middleware platforms that can embed native evaluation hooks directly into their execution runtimes. 3. **Agent Infrastructure Teams (e.g., Harness)**: Core infrastructure providers who can offer standardized logging, tracing, and eval data pipelines at the foundational layer.

Team Bio

Part of the Responsible AI team.

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Background

Why are evals important?

1. **Governance drives delivery**: When guardrails are clear and automated, teams build with confidence. When policies travel with the code, security reviews become approvals instead of interrogations. When compliance is infrastructure rather than inspection, pilots graduate to production in weeks, not quarters.
2. **Evaluating agentic systems are challenging**: you need to consider non-deterministic execution paths, compounding step errors, state drift, and tool-invocation failures.

Core Terminology

1. **Observability**: Passive capture of live system telemetry (traces, logs, spans).
2. **Evaluations (Evals)**: Active execution of assertions, trajectory scoring, and guardrail checks against test sets or live traces.
3. **Benchmarking**: Comparative performance scoring against static datasets or state-of-the-art (SOTA) baselines.

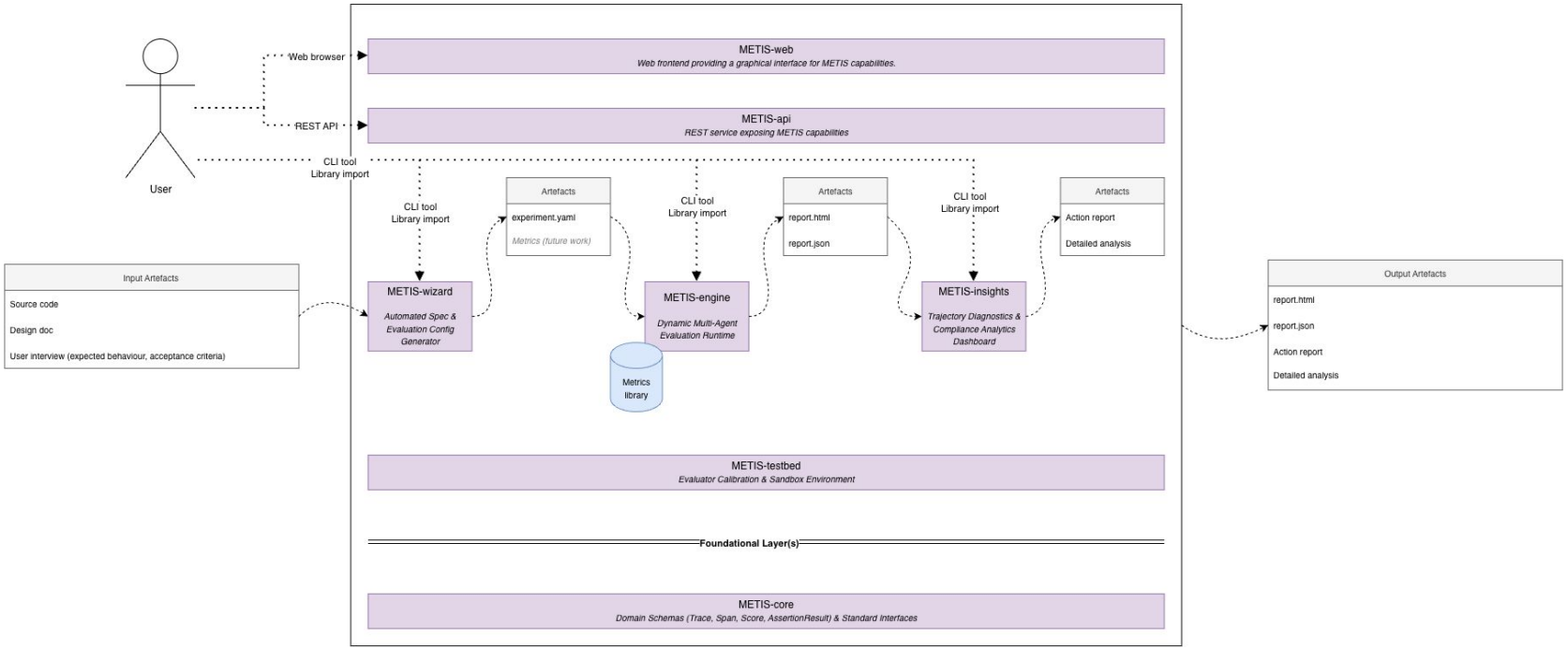
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Project Management

METIS Product Suite

A unified product suite that delivers end-to-end evaluation, from configuration to actionable insights (src).



Product	Role	Key Function
METIS	METIS presents a single, consistent user interface designed for Whole-of-Government adoption—abstracting backend complexity into a seamless workflow.	
METIS-wizard	The front-facing interface for defining and configuring evaluations.	Guides project teams through setting up evaluation test sets, trajectory assertions, and guardrails with minimal friction—eliminating boilerplate code and manual setup.
METIS-engine	The high-performance backend evaluation engine.	Ingests telemetry and trace data across diverse agentic architectures and formats (e.g., Langfuse traces). It handles the heavy lifting of non-deterministic trajectory scoring, step-error analysis, and automated guardrail checks.
METIS-insights	The reporting and decision-support dashboard.	Translates raw trace evaluations into high-level risk profiles, benchmark scores, and drill-down failure diagnostics—giving both developers and agency leadership audit-ready safety and reliability metrics.
METIS-testbed	The isolated testing ground for automated scenario execution and stress-testing.	Provides sandboxed environments, mock tool/API integrations, and curated benchmark datasets—enabling teams to simulate complex, multi-turn agent interactions, adversarial edge cases, and safety attacks without impacting live production systems.

Product	Description	Positioning	Role in Lifecycle
METIS	METIS presents a single, consistent user interface designed for Whole-of-Government adoption—abstracting backend complexity into a seamless workflow.		
METIS-engine	An evaluation tool for AI agent systems that interacts with the live application - simulating realistic user personas, driving multi-turn conversations through the actual product interface, and scoring every agent in the chain. Orchestration logic, delegation decisions, and downstream execution are all evaluated in context, as a single integrated system.	The core execution engine that runs multi-turn persona simulations, trajectory assertions, and intermediate chain evaluations against live application runtimes and trace streams.	Replaces static single-prompt benchmarks with interactive, end-to-end system testing (evaluating orchestration, tool invocation, and multi-agent handoffs in context).
METIS-wizard	Automatically extracts evaluation targets from source code and PRD to generate task.yaml and persona.yaml.	A developer onboarding/product discovery CLI/SDK that parses source code, PRDs, and policy rules to automatically scaffold evaluation datasets, task parameters, and threat personas (task.yaml, persona.yaml).	Eliminates the manual burden of writing test suites, drastically lowering setup friction for engineering teams.
METIS-insights	Pinpoints failure modes across prompt iterations or underlying model updates.	The post-execution analytics engine that performs failure-mode root-cause analysis, tracks evaluation regressions across CI/CD builds, and generates government safety and alignment compliance reports.	Translates raw trace-level evaluation scores into actionable debugging paths for developers and audit sign-offs for team leads.
METIS-testbed	A local, containerized testbed for evaluating AI agents and agentic systems and ensuring relative scoring consistency across open-source evaluation engines.	An isolated, containerized harness designed for meta-evaluation—benchmarking evaluator precision, calibrating LLM-as-a-judge scoring consistency, and validating open-source eval frameworks before upstream deployment.	Serves as the "eval for evals" tool to guarantee low metric variance, prevent judge bias, and benchmark platform updates.

Milestones

2026-Q2

Phase 1: *Exploration*

1. Literature survey
2. Small-scale experimentation

2026-08

Phase 2: *METIS MVP*

1. DDL: 2026-08-12 (sharing with management)
2. Deliverables
   a. METIS-engine v0 (ingest Langfuse traces)
   b. METIS-wizard v0
3. Customized to meet the evaluation requirements for ORCA (MOM)

2026-09

Phase 3: Generalized METIS

1. Deliverables
   a. METIS-engine v1 (ingest Langfuse traces)
   b. METIS-wizard v1
2. Potential users
   a. Harness (AI Practice)

2026-10

Phase 4: Expand Integration Targets

1. Deliverables
   a. METIS-engine v2 (support different trace formats and agentic architectures)
   b. METIS-insights v1
2. Potential users
   a. ATLAS (ESG)

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Epilogue

References

1. A Primer on Evals by @Shaun_KHOO

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Thank you.

CLASSIFICATION